# Engineering Standard - Bolted Joints and Tightening Methods

| | | | |
|---|---|---|---|
| **Document no.** | ES-TOR-001 | **Revision** | D |
| **Document type** | Engineering Standard | **Status** | Current |
| **Effective date** | 2025-10-01 | **Owner** | Product Engineering |
| **Classification** | Internal | | |

## 1 Scope

This standard applies to all bolted joints in Nordwerk e-drive products. It defines joint classes, tightening methods and the rules for reuse of fasteners.

## 2 Joint classes

| Class | Description | Example | Tightening method |
|---|---|---|---|
| A | Safety critical, failure can cause injury | Mounting bolts e-drive to subframe | Torque-angle with monitoring, 100 % documented |
| B | Function critical, failure causes breakdown | Gearbox flange, cover bolts | Torque controlled with angle monitoring, 100 % documented |
| C | Other joints | Brackets, covers without sealing | Torque controlled |

## 3 Tolerances

The default torque tolerance for class B and C joints is +/- 10 % of the nominal torque unless the drawing says otherwise. For class A joints the tolerance is defined in the drawing.

## 4 Reuse of fasteners

- Bolts with pre-applied threadlocker or microencapsulated coating must never be reused.
- Torque-to-yield bolts (class A) must never be reused.
- Other bolts may be reused up to 2 times if the thread is undamaged.
- Self-locking nuts must always be replaced.

## 5 Torque tool capability

Production torque tools must have a machine capability Cmk of at least 1.67. Tools are checked with the torque analyser at shift start and fully calibrated every 12 months.

## 6 Revision history

| Rev | Date | Change |
|---|---|---|
| C | 2024-08-20 | Joint classes revised |
| D | 2025-10-01 | Cmk requirement 1.33 -> 1.67 |